Agniva Maiti

+91 87298 06726 | maitiagniva@gmail.com | Dimapur, India | linkedin.com/in/agniva-maiti | github.com/AgnivaMaiti | agniva.in

Summary

ML-focused Computer Science graduate bridging advanced machine learning and scalable software architecture, with papers at EMNLP 2025 (Main), WMT 2026 and GlobalSouthML at ICML 2026. Experienced in training, fine-tuning (LoRA/QLoRA) and deploying deep learning models (LLMs, NLP, computer vision) with PyTorch and TensorFlow, rapid prototyping, and integrating ML pipelines (NLP, LLM agents) into production using agile methodologies.

Education

Kalinga Institute of Industrial Technology (KIIT), Bhubaneswar Sep 2022 - May 2026
Bachelor of Technology (B.Tech) in Computer Science and Engineering *CGPA: 8.72 / 10.0*

- **Relevant Coursework:** Machine Learning, GenAI & Large Language Models, Deep Learning, Data Structures & Algorithms, Probability & Statistics, Data Mining, Discrete Math, DBMS, Software Engineering, OS, Computer Networks

Skills

Languages: Python, Java, C, C++, JavaScript, SQL, Dart | **Methods**: Data Science, Statistical Analysis, Agile
AI/ML: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, Pandas, NumPy, LLMs, Generative AI, Agentic AI, NLP
ML Techniques: RAG, LoRA/QLoRA Fine-Tuning, Prompt Engineering, Machine Translation, Computer Vision, Image Processing
ML Deployment: TensorFlow Lite, Model Quantization, On-Device Inference, ML Inference Pipelines, LLM Evaluation
Development: Android Development (Flutter, React Native), React, AWS, Git, GitHub, Docker, PostgreSQL, Firebase, Flask

Experience

Freelance | **Speech Recognition (ASR)** Aug 2026 - Sep 2026

- Fine-tuned, aligned and benchmarked automatic speech recognition (ASR) models for Yiddish and Hebrew.

DES Navalmal Firodia Law College (Agrovigya) | **App Development Intern** Feb 2025 - Jul 2025

- Developed the Agrovigya Flutter app serving 36 districts by integrating quantized TensorFlow Lite models for offline use, and engineered Random Forest classifiers (Scikit-learn) to power the website's crop and job prediction backend.

IIT Dhanbad (Dept. of Mathematics and Computing) | **Research Intern** Dec 2024 - Jan 2025

- Reviewed existing literature (20+ papers) and reproduced and benchmarked 5+ SOTA low-light enhancement models (e.g., Zero-DCE, DVN) using SSIM/PSNR metrics, optimizing pre-processing pipelines for improved restoration.

IIT Kharagpur (Dept. of Artificial Intelligence) | **Research Intern** May 2024 - Jul 2024

- Curated a large-scale benchmark dataset of 18,700+ entries by aggregating and annotating query results from Google and Bing to analyze search engine bias.
- Leveraged 4 Small Language Models (SLMs, like Mistral and Phi) to evaluate information retrieval fairness, quantifying relevance and stance across diverse search queries.

RnPsoft Pvt. Ltd. | **Machine Learning Intern** Jan 2024 - Aug 2024

- Architected scalable, end-to-end ML inference pipelines for personalized educational media delivery, serving 500+ users.

Projects

NagaNLP | ***Python, PyTorch, Transformers, Low-Rank Adaptation, Low-Resource NLP*** | **arXiv** | **HuggingFace**

- Formulated a novel Human-in-the-Loop bootstrapping pipeline to synthesize a 10,018-pair conversational instruction corpus and annotated POS/NER datasets, with inter-annotator agreement of Cohen's kappa 0.92 (POS) and 0.88 (NER).
- Fine-tuned NagaLLaMA (Llama-3.2-3B) yielding 3.85 perplexity (vs. 96.76 few-shot baseline) and transformer taggers with 93.81% POS accuracy and 0.75 NER F1; released as the first open-source NLP toolkit for Nagamese (4 models, 3 datasets).

NagaTranslate | ***React, Next.js, Machine Translation, Multilingual NLP*** | **Live App**

- Developed a scalable web application for 3 low-resource indigenous languages (Nagamese, Ao, Sema), now with 4 more in beta, engineering a text-routing pipeline to digitize cross-cultural dialogue for 2M+ regional speakers.

GScholar App | ***Flutter, Dart, LLMs, AdMob*** | **Play Store**

- Architected a production-grade webview Flutter application, scaling to 100,000+ downloads (4.3 rating); integrated LLM-driven tools including ELI5 document chat and multi-paper comparative analysis while optimizing AdMob monetization pipelines.

Publications & Workshops

Investigating Pedagogical Teacher and Student LLM Agents | ***EMNLP 2025 (Main), 2nd author*** | **ACL Anthology**

- Engineered a multi-agent framework using Genetic Algorithms to evolve teacher strategies over 50 generations; scaled to 100 heterogeneous student agents (final mean score 8.14/10).
- Proposed Persona-RAG, a cognitive retrieval module achieving 88% accuracy on conceptual questions (vs. 85% HyDE, 74% vanilla RAG); human evaluation with 20 raters (inter-rater reliability: Krippendorff's alpha 0.78-0.84).

Bridging the Visual Gap: Resource-Efficient VLM Adaptation for Meitei Mayek using Synthetic Multi-Modal Mixtures | ***GlobalSouthML, ICML 2026*** | **Paper**

- Engineered a 5-piece multimodal synthetic data mixture to adapt PaliGemma-3B to the Meitei Mayek script via QLoRA.
- Raised ANLS 8.62% to 21.62%, doubled VQA Exact Match (8.00% to 16.00%) and reached 76.0% POPE accuracy (85.36% F1).

Evaluating Indic Literary Creativity In LLMs through the Lens of the Shastras | ***GlobalSouthML, ICML 2026*** | **Paper**

- Benchmarked 8 leading LLMs across 2 Indic languages (Hindi, Bengali) by adapting 4 dimensions of the Torrance Tests of Creative Thinking with 6 canonical rhetorical devices.
- Uncovered an algorithmic Alankara over-stuffing bias via an ICC-validated LLM-as-a-judge pipeline, exposing critical inflation vulnerabilities across density-based evaluation heuristics.

SCE-KIIT: Fine-Tuning NLLB-200 for English-Nagamese Translation | ***WMT 2026*** | **Paper**

- Curated 2,280 pairs and fine-tuned NLLB-200 (600M) with surrogate language tokens; internal held-out BLEU 0.40 to 40.02.

SCE-KIIT: KokLLaMA: Cross-Task Adaptation for Low-Resource English-Kokborok Translation | ***WMT 2026*** | **Paper**

- QLoRA-tuned Llama-3.2-3B-Instruct on 11,428 Kokborok instruction pairs, with no parallel training data.

Leadership & Activities

KIIT Robotics Society & KIIT Electrical Society | App Development Lead: Led and mentored 100+ students across both societies, spearheading Flutter-based development of 10+ production-grade applications.

KIITFest 8.0 | Volunteer App Developer: Deployed an internal mobile scanning tool enabling volunteers to verify IDs and manage merchandise distribution for 1,500+ attendees, reducing entry queue downtime by 40%.

NeurIPS 2026 GlobalSouthAI Workshop | Reviewer: Reviewed workshop paper submissions (Sep 2026).